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Division: A2

Subject: Programming with Java

ASSIGNMENT No : 9

TITLE :Write a Java program to demonstrate the use of the final keyword with variables, methods, and classes.

Problem Statement

Write a Java program to demonstrate the use of the final keyword with variables, methods, and classes.

Learning Objectives

To implement the final keyword with variables, methods, and classes.

Theory

The final keyword restricts modification in Java programs. A final variable cannot be reassigned after

initialization. A final method cannot be overridden by subclasses, while a final class cannot be inherited. This

improves security, prevents accidental modifications, and increases program reliability. Final is commonly

used for constants and immutable classes.

Outcome

Conclusion

These programs demonstrate the use of the final **keyword** to create values that cannot be changed after initialization, ensuring data remains constant and secure

Exercise

1. Create a Bank Account program where account number is final and cannot be changed.

```
import java.util.Scanner;
```

```
class BankAccount {
```

```
    final int accountNo;
```

```
    String name;
```

```
    double balance;
```

```
    BankAccount(int accountNo, String name, double balance) {
```

```
        this.accountNo = accountNo;
```

```
        this.name = name;
```

```
        this.balance = balance;
```

```
    }
```

```
    void display() {
```

```
        System.out.println("Account Number: " + accountNo);
```

```
        System.out.println("Name: " + name);
```

```
        System.out.println("Balance: " + balance);
```

```
    }
```

```
}
```

```
public class Main {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter Account Number: ");
```

```
        int accNo = sc.nextInt();
```

```
        sc.nextLine();
```

```
System.out.print("Enter Name: ");
```

```
String name = sc.nextLine();
```

```
System.out.print("Enter Balance: ");
```

```
double balance = sc.nextDouble();
```

```
BankAccount b = new BankAccount(accNo, name, balance);
```

```
b.display();
```

```
}
```

```
}
```

2. Create a Library Book Management program where the book ISBN is declared as final and cannot be changed once assigned. Display the book's ISBN, title, author, and price.

```
import java.util.Scanner;
```

```
class Book {
```

```
    final String isbn;
```

```
    String title;
```

```
    String author;
```

```
    double price;
```

```
    Book(String isbn, String title, String author, double price) {
```

```
        this.isbn = isbn;
```

```
        this.title = title;
```

```
        this.author = author;
```

```
        this.price = price;
```

```
    }
```

```
    void display() {
```

```
        System.out.println("ISBN: " + isbn);
        System.out.println("Title: " + title);
        System.out.println("Author: " + author);
        System.out.println("Price: " + price);
    }
}
```

```
public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter ISBN: ");
        String isbn = sc.nextLine();

        System.out.print("Enter Title: ");
        String title = sc.nextLine();

        System.out.print("Enter Author: ");
        String author = sc.nextLine();

        System.out.print("Enter Price: ");
        double price = sc.nextDouble();

        Book b = new Book(isbn, title, author, price);
        b.display();
    }
}
```